INCH-POUND
A-A-51140B
25 September 2000
SUPERSEDING
A-A-51140A
23 March 1990

COMMERCIAL ITEM DESCRIPTION

CHUCKS, DRILL, KEY TYPE, THREE-JAW, GEARED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for geared three-jaw drill chucks capable of being adjusted to hold any size straight shank tool within the capacity of the chuck. The jaws of the chuck shall be opened and closed by means of a pinion-type key.

2. CLASSIFICATION. The drill chucks shall be classified and identified by the types (see 7.3(b)), styles (see 7.3(c)), and sizes listed below. Unless otherwise specified in the acquisition order (see 7.3(d)), the drill chuck size shall be one of the coded options listed in tables I, II, or III.

2.1 Type I. Tapered-mount

Style A - medium duty chucks - intended for use on stationary and portable power tools.

Size – see table I

Style B - heavy duty chucks - suitable for use with heavy-duty portable power tools.

Size – see table II

Style C - extra heavy duty chucks - shall be used with production drilling machines, jig borers, radial drill presses, milling machines, and lathes.

Size – see table III

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3460

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2.2 Type II. Threaded-mount

Style A - medium duty chucks - intended for use on stationary and portable power tools.

Size – see table I

Style B - heavy duty chucks - suitable for use with heavy-duty portable power tools.

Size – see table II

3. SALIENT CHARACTERISTICS

3.1 Material. The chuck body shall be constructed of carbon steel or tool steel. All chuck jaws shall be made of alloy steel. Chuck keys shall be made of cold rolled steel or equivalent.

3.2 Hardness. The sides of the key holes shall be hardened to not less than 48 on the Rockwell "C" scale when measured in accordance with the American Society for Testing and Materials (ASTM) E 18, "Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials" (DoD adopted). The gripping surface of each chuck jaw shall have a hardness of not less than 52 and each chuck key shall be hardened on the bearing surface and shank to a value of not less than 42 on the Rockwell "C" scale when measured in accordance with ASTM E 18.

3.3 Tapered-mount hole chuck. The tapered-mount hole of type 1 chucks shall conform to the requirements of table I, II, or III for the chuck size specified.

3.4 Threaded-mount hole chuck. The threaded-mount hole of type 2 chucks shall conform to the dimensions of tables I or II for the chuck size specified. The threads shall conform to type 2B tolerance requirements of the American Society of Mechanical Engineers (ASME) Standard B1.1, "Unified Inch Screw Threads (UN and UNR Thread Form)" (DoD adopted). The direction of the threads shall be such that to remove the chuck from a tool spindle, the chuck must be turned in the same direction the spindle rotates when in use. Chucks designed for use on reversible tools shall have a retaining screw or other suitable means to secure the chuck during reverse operations.

3.5 Bearings. All style A and B drill chucks shall have plain bearings. Style C drill chucks shall contain ball thrust bearings. The ball thrust bearings shall be of a standard commercial quality, readily available for replacement purposes. If thrust bearings are not sealed-for-life, means shall be provided for lubrication.

3.6 Drill chuck capacity. The drill chuck shall conform to the drill holding capacity listed in the respective tables for the type and style chuck required.

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TABLE I. Type I and II, style A chucks.

Size code	Capacity range (inches)	Mount hole end of chuck		Chuck dimensions (inches)	
		Jacobs taper number type 1, style A	Nominal diameter (inches) and thread, type 2, style A	Overall length closed	Sleeve diameter
01	0 - 5/32	0	5/16 x 24	1-3/8	53-64
02	1/64 - 5/32	0	----	1-45/64	1
03	0 - 1/4	----	5/16 x 24	1-15/16	1-3/16
04	0 - 1/4	1	3/8 x 24	1-15/16	1-1/8
05	0 - 1/4	----	1/2 x 20	2-1/4	1-5/16
06	1/64 - 1/4	1	*3/8 x 24	2-1/16	1-7/64
07	1/64 - 1/4	----	1/2 x 20	2-23/64	1-9/32
08	1/64 - 5/16	1	*3/8 x 24	2-19/64	1-9/32
09	1/64 - 5/16	----	1/2 x 20	2-23/64	1-9/32
10	0 - 3/8	2	*3/8 x 24	2-3/4	1-11/16
11	0 - 3/8	----	1/2 x 20	2-3/4	1-25/32
12	1/64 - 3/8	----	3/8 x 24	2-23/64	1-19/64
13	1/64 - 3/8	----	1/2 x 20	2-23/64	1-19/64
14	1/32 - 3/8	2	*3/8 x 24	2-7/8	1-5/8
15	1/32 - 3/8	33	1/2 x 20	2-7/8	1-5/8
16	1/16 - 3/8	----	3/8 x 24	2-13/32	1-27/64
17	1/16 - 3/8	----	*1/2 x 20	2-13/32	1-27/64
18	0 - 1/2	2	1/2 x 20	3-7/16	1-25/32
19	0 - 1/2	33	5/8 x 16	3-5/64	1-25/32
20	0 - 1/2	6	----	3-7/16	2-5/64
21	1/32 - 1/2	2	1/2 x 20	3-3/8	2
22	1/32 - 1/2	6	5/8 x 16	3-3/8	2
23	1/32 - 1/2	33	----	3-3/8	2
24	1/16 - 1/2	----	*3/8 x 24	2-7/8	1-5/8
25	1/16 - 1/2	----	*1/2 x 20	2-7/8	1-5/8
26	5/64 - 1/2	33	3/8 x 24	3-3/32	1-25/32
27	5/64 - 1/2	----	*1/2 x 20	3-3/32	1-13/16
28	5/64 - 1/2	----	*5/8 x 16	3-3/32	1-13/16
29	1/32 - 5/8	3	5/8 x 16	3-55/64	2-7/32
30	1/8 - 5/8	3	5/8 x 16	3-13/16	2-19/64
31	1/8 - 5/8	----	3/4 x 16	3-13/16	2-3/8
32	5/32 - 3/4	3	----	3-55/64	2-7/32
33	3/16 - 3/4	3	5/8 x 16	3-13/16	2-3/8
34	3/16 - 3/4	4	3/4 x 16	3-57/64	2-3/8
35	1/4 - 13/16	3	----	3-21/64	2-33/64
36	3/16 - 1-1/32	5	----	5-25/32	3-7/64

*For use on reversible tools.

NOTE: All dimensions shall not be less than those shown. Where a capacity range is shown, the dimensions shall not be less than the lowest figure to not greater than the highest figure.

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TABLE II. Type I and II, style B chucks.

Size code	Capacity range (inches)	Mount hole end of chuck		Chuck dimensions (inches)	
		Jacobs taper number type 1, style B	Nominal diameter (inches) and thread, type 2, style B	Overall length closed	Sleeve diameter
01	0 - 1/4	1	*3/8 x 24	2-7/32	1-19/64
02	0 - 1/4	2 short	1/2 x 20	2-7/32	1-19/64
03	0 - 5/16	2 short	3/8 x 24	2-5/16	1-27/64
04	0 - 5/16	----	1/2 x 20	2-13/32	1-27/64
05	0 - 3/8	2	*1/2 x 20	2-3/4	1-25/32
06	0 - 3/8	----	5/8 x 16	2-3/4	1-25/32
07	0 - 1/2	6	5/8 x 16	3-7/16	2-1/16
08	0 - 1/2	----	45/64 x 16	3-17/32	2-1/16
09	0 - 1/2	----	3/4 x 16	3-17/32	2-1/16
10	1/8 - 5/8	3	5/8 x 16	3-13/16	2-5/16
11	3/16 - 3/4	----	5/8 x 16	3-1/64	2-3/8
12	3/16 - 3/4	3	3/4 x 16	3-57/64	2-3/8
13	1/4 - 13/16	3	----	4-11/32	2-9/16

*For use on reversible tools.
NOTE: All dimensions shall not be less than those shown. Where a capacity range is shown, the dimensions shall not be less than the lowest figure to not greater than the highest figure.

TABLE III. Type I, style C chucks.

Size code	Capacity range (inches)	Jacobs taper number type 1, style C	Chuck dimensions (inches)	
			Overall length closed	Sleeve diameter
01	0 - 1/4	2 short	2-11/32	1-9/16
02	0 - 3/8	2	2-45/64	1-49/64
03	0 - 1/2	3	3-27/32	2-23/64
04	1/32 - 1/2	3	3-53/64	2-15/64
05	1/8 - 5/8	3	4-5/16	2-5/8
06	1/32 - 5/8	3	3-57/64	2-7/16
07	1/8 - 3/4	4	4-57/64	2-53/64
08	3/16 - 3/4	4	4-41/64	2-51/64
09	3/8 - 1	5	5-1/2	3-21/32
10	3/16 - 1-1/32	5	4-23/32	3-7/64

NOTE: All dimensions shall not be less than those shown. Where a capacity range is shown, the dimensions shall not be less than the lowest figure to not greater than the highest figure.

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3.7 Chuck key. Unless specified in the acquisition order (see 7.3(e)), each key operated chuck shall be furnished with a standard pinion type key. When specified in the acquisition order (see 7.3(f)), a self-ejecting safety key shall be furnished. The pinion key shall mesh with and operate smoothly with the geared portion of the chuck for 360 degrees of rotation. Each key handle shall have sufficient diameter and leverage to lock the tool firmly in the chuck without binding or breaking the handle. The handle may be a steel pin either rounded on both ends or flattened at one end for application of additional thumb pressure. Self-ejecting keys shall have a built-in mechanism that shall eject the key from the chuck body when hand pressure upon the key is relieved.

3.8 Chuck runout and accuracy. When mounted on a true arbor, the plain bearing chuck shall run concentric within 0.004 inch total indicator reading (TIR) on any outside diameter of the chuck body. The maximum chuck runout shall be 0.004 inch TIR when checked with two true test plugs alternately tightened within the jaws, one having a diameter equal to one half the maximum capacity of the chuck and the other having the maximum capacity. The runout shall be checked 1-1/2 inches from the end of the chuck body on chuck of 3/8 inch capacity and larger. The runout shall be checked at one inch from the end of the chuck body on chucks 1/4 to 5/16 inch capacity. The runout shall be checked 3/4 inch from the chuck body on chucks having 5/32 inch capacity. The maximum runout for ball bearing chucks shall be 0.003 inch TIR when checked one inch from the chuck nose.

3.9 Workmanship. Each drill chuck and key shall be free from any imperfections or defects that may impair serviceability, durability, or appearance.

3.10 Markings. Unless specified otherwise in the acquisition order (see 7.3(g)), each drill chuck shall be plainly marked with the manufacturer's name, model number, and capacity.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.3(h)).

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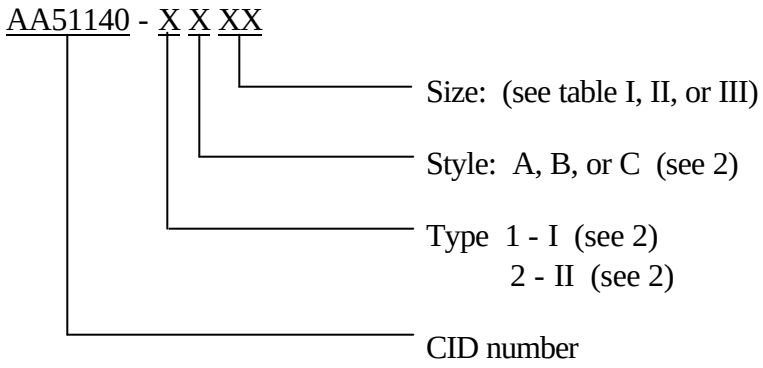
7. NOTES

7.1 Part or identification number (PIN). The following part or identification numbering system is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.2.2 ASME standards. Copies of ASME standards may be obtained from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

7.2.3 ASTM standards. Copies of ASTM standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.3 Ordering data. Acquisition documents should specify the following information:

a. CID document number, revision, and CID PIN
b. Chuck type (see 2)
c. Chuck style (see 2)
d. Chuck size (see 2)
e. Chuck key, if different (see 3.7)
f. Self-ejecting chuck key, if required (see 3.7)
g. Markings, if different (see 3.10)
h. Packaging, (see 6.1)

7.4 National stock number (NSN). The NSNs listed in table IV are assigned for the listed product types and sizes (see table I, II, and III). Other NSNs may also correspond with this document.

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TABLE IV. Assigned NSNs.

Type	Style	Size	Assigned NSN
1	A	4	3460-00-187-2199
1	A	4	3460-00-231-2251
1	A	7	3460-00-231-2261
1	A	18	3460-00-231-2253
1	A	19	3460-00-189-4614
1	B	7	3460-00-969-5313
1	B	12	3460-00-249-2842
1	C	1	3460-00-268-9218
1	C	2	3460-00-268-9219
1	C	3	3460-01-213-1755
1	C	3	3460-00-231-2256
1	C	3	3460-00-268-9220
1	C	7	3460-00-231-2258
1	C	7	3460-00-268-9221
1	C	7	3460-00-540-1084
1	C	9	3460-00-268-9222
1	C	9	3460-00-516-4597
2	A	10	3460-01-048-3039
2	A	10	3460-00-189-4617
2	A	27	3460-00-189-4624
2	B	1	3460-00-277-8297
2	B	5	3460-00-189-4618
2	B	7	3460-00-189-4621
2	B	7	3460-00-270-6021
2	B	9	3460-00-489-2106
2	B	11	3460-00-189-4626

MILITARY INTERESTS:

Custodians:
Army - AL
Air Force - 99

Reviewer:
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

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